

TESTLEAF TRAINING

Playwright

XPATH-Axes



How? When?

The Golden Rule

What is Xpath-Axes?

XML path based relationship

When to use?

Not able to locate unique WebElement using Basic xpath.

How to write xpath?

In relation with other elements

Xpath (RelationShip based)

LeafTaps- Login page

Let's Analyse the Relationship

<div> represents the ancestor or grandparent

<form> represents parent to <p> and child of <div>

<p> represents child to <form>

Parent-child

siblings

cousin

<label>

<input>

Xpath-Axes (type 1 of 8)

Parent to child :

To reach the WebElement from <form>(parent tag) to <p> (child)

Syntax:

```
( parent relative xpath)/tagName of child
```

Example:

```
//form[@id='login']/p
```

Xpath-Axes (type 2 of 8)

Contd..

GrandParent to child relation :

To reach the WebElement from <form>(grandparent tag) to <label> (grand-child)

Syntax:

```
GrandParent relative xpath/tagName of grand child
```

Example:

```
(//form[@id='login'])//label
```

Xpath-Axes (type 3 of 8)

Contd..

Child to parent :

To reach the WebElement from <p>(child) to <form> (parent)

Syntax:

```
child relative xpath/parent::parent tagName
```

Example:

```
//input[@id='username']/parent::p
```

Xpath-Axes (type 4 of 8)

Contd..

GrandChild to Grand parent :

To reach the WebElement from <label>(grand-child) to <form> (grand-parent), we have the keyword ancestor

Syntax:

```
GrandChild relative xpath/ancestor::tagName of grand parent
```

Example:

```
//input[@id='password']/ancestor::form
```

Xpath-Axes (type 5 of 8)

Contd..

Elder Sibling to Younger Sibling :

To reach the WebElement from <label> (child) to <label> (child) of same parent.

we have the keyword following-sibling (finding from top to bottom)

Syntax:

(Elder sibling relative xpath)/following-sibling::tagName of younger sibling

Example:

```
//label[text()='Username']/following-sibling::input
```

Xpath-Axes (type 6 of 8)

Contd..

Younger Sibling to elder Sibling:

To reach the WebElement from Second <label> (child) to first <label> (child) of same parent we have the keyword preceding-sibling (finding from bottom to top)

Syntax:

```
Younger sibling relative xpath/preceding-sibling::tagName of Elder sibling
```

Example

```
//input[@id='username']/preceding-sibling::label
```

Xpath-Axes (type 7 of 8)

Contd..

Elder cousin to Younger cousin :

To reach the WebElement from first parent child <label> (child) to second parent child <label> (child) we have the keyword following (finding from top to bottom)

Syntax:

```
Elder cousins relative xpath/following::tagName of younger Cousin
```

Example

```
//input[@id='username']/preceding-sibling::label
```

Xpath-Axes (type 8 of 8)

Contd..

Younger cousin to Elder cousin :

To reach the WebElement from second parent child <label> (child) to first parent child <label> (child) we have the keyword preceding (finding from bottom to top)

Syntax:

```
Younger cousin xpath/preceding::tagName of Elder cousin
```

Example:

```
//label[text()='Password']/preceding::input
```

Quick Recall

✓ Axes Xpath-What , when, How?

✓ 8 types of Relationship based Xpath

■ Parent to Child & Viceversa

■ Grandparent to Grandchild & Viceversa

■ Sibling:

Elder to younger

Younger to Elder

■ Cousin

Elder to younger

Younger to Elder